Android AlarmManager

Android AlarmManager allows you to access system alarm.
By the help of **Android AlarmManager** in android, you can *schedule your application to run at a specific time* in the future. It works whether your phone is running or not.

The Android AlarmManager holds a CPU wake lock that *provides guarantee not to sleep the phone* until broadcast is handled.

Android AlarmManager Example
Let's see a simple AlarmManager example that runs after a specific time provided by user.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start"
        android:layout_alignParentBottom="true"
        android:layout_centerHorizontal="true"
        android:layout_marginBottom="103dp" />

    <EditText
        android:id="@+id/time"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentTop="true"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="22dp"
        android:ems="10" />
</RelativeLayout>
```

MainActivity.java

The activity class starts the alarm service when user clicks on the button.

```
import android.app.AlarmManager;
import android.app.PendingIntent;
import android.content.Intent;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends Activity {
   Button start;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      start= findViewById(R.id.button);

      start.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View view) {
            startAlert();
         }
      });
   }

   public void startAlert(){
      EditText text = findViewById(R.id.time);
      int i = Integer.parseInt(text.getText().toString());
      Intent intent = new Intent(this, MyBroadcastReceiver.class);
      PendingIntent pendingIntent = PendingIntent.getBroadcast(
            this.getApplicationContext(), 234324243, intent, 0);
      AlarmManager alarmManager = (AlarmManager) getSystemService(ALARM_SERVICE);

      alarmManager.set(AlarmManager.RTC_WAKEUP, System.currentTimeMillis()
            + (i * 1000), pendingIntent);
      Toast.makeText(this, "Alarm set in " + i + " seconds",Toast.LENGTH_LONG).show();
   }
}
```

Let's create BroadcastReceiver class that starts alarm.

File: MyBroadcastReceiver.java

```
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.media.MediaPlayer;
import android.widget.Toast;

public class MyBroadcastReceiver extends BroadcastReceiver {
    MediaPlayer mp;
    @Override
    public void onReceive(Context context, Intent intent) {
        mp=MediaPlayer.create(context, R.raw.alarm);
        mp.start();
        Toast.makeText(context, "Alarm....", Toast.LENGTH_LONG).show();
    }
}
```

File: AndroidManifest.xml

You need to provide a receiver entry in AndroidManifest.xml file.

```
<receiver android:name=".MyBroadcastReceiver" ></receiver>
```